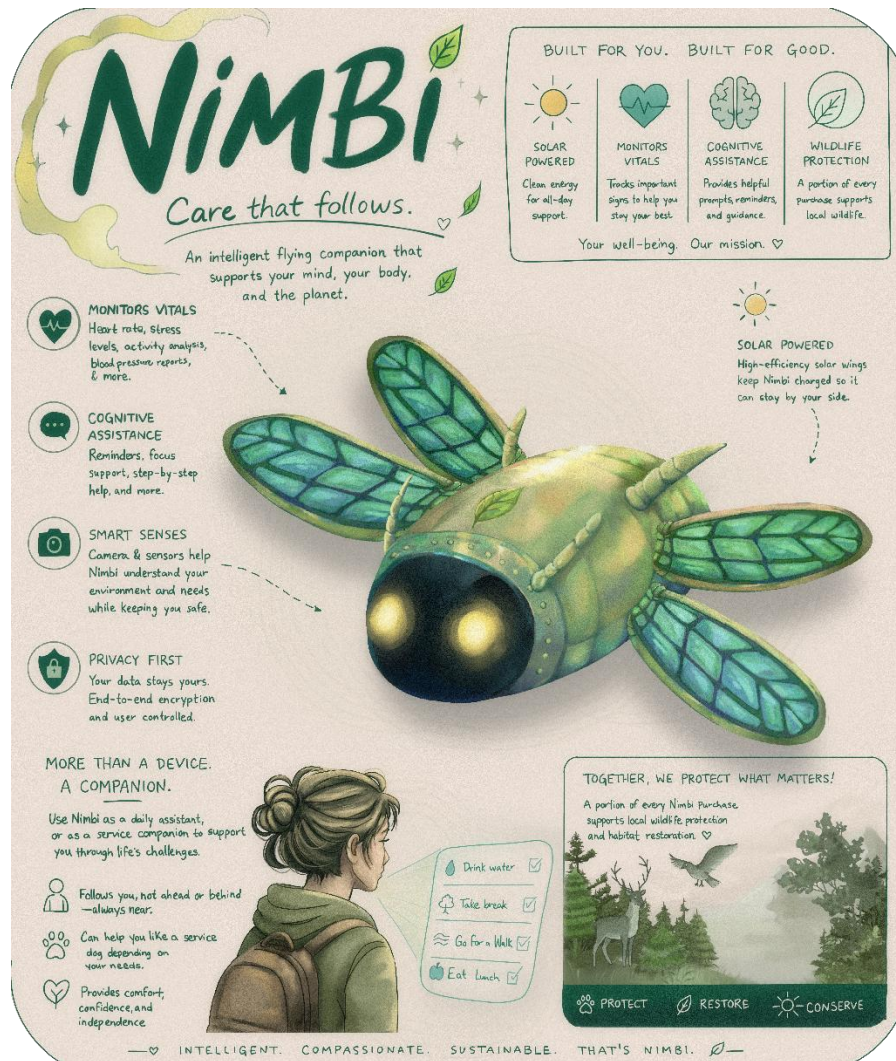


NIMBI

Care That Follows

Digital Exhibit Entry | Tatiana Choate

TCID — Augmentation Prototype & Digital Exhibit



Overview

Nimbi is a speculative companion drone designed to follow its user through daily life — monitoring vitals, delivering customizable reminders, and providing a consistent presence for people whose bodies and minds do not always send reliable signals. Part assistive technology, and part emotional support system, Nimbi explores what it might mean for care to become mobile, adaptive, and continuously present.

Rather than asking users to constantly manage themselves, Nimbi imagines a form of augmentation that follows the user and responds to human needs in real time.

What It Augments

Nimbi augments the body's internal signaling systems: the automatic processes people often rely on without thinking. Such as, remembering to eat, noticing dehydration, recognizing a drop in blood pressure, or realizing that hours have passed without movement are tasks many people perform instinctively. For users whose cognitive or physical conditions disrupt internal awareness, those signals can become inconsistent, delayed, or difficult to interpret.

Nimbi is designed to help users recognize those needs. Through reminders, health monitoring, and pattern recognition, the drone extends the user's ability to track and respond to their own needs without replacing their agency or independence.

Who It Is For

Nimbi was designed with accessibility-centered users in mind. Its primary audience includes people who have trouble with routine self-monitoring, memory, energy management, or recognizing physical and cognitive warning signs in daily life. Rather than focusing on a single diagnosis, the project considers the broader experience of navigating a body or mind that does not always communicate its needs consistently.

The project also considers users who travel or live alone and may benefit from a persistent monitoring presence capable of detecting distress or emergencies; or seniors experiencing isolation, caregivers coordinating complex routines, and individuals whose demanding schedules interrupt basic self-care practices could also benefit from Nimbi.

At its core, Nimbi declares that care should follow and help the user rather than require the user to constantly seek it out or log it themselves.

How It Works

Nimbi is imagined as a small bug-inspired companion drone that hovers near its user throughout the day. Its rounded robotic body and leaf-shaped wings were intentionally designed to feel approachable rather than clinical, through blending soft sage colors and natural forms with visible technology. The drone's wings function as solar panels, allowing it to charge passively outdoors — a practical response to the reality that users with memory impairments or executive dysfunction may forget to charge a traditional device consistently.

As it follows the user, Nimbi continuously monitors patterns in heart rate, blood pressure, stress, movement, and activity. Rather than operating as a purely reactive alert system, the device would use AI-supported pattern recognition to identify changes in routine and deliver

proactive reminders for hydration, medication, meals, movement, or rest before small problems escalate into larger ones.

Its facial camera system would provide wide-angle environmental awareness that supports both navigation and emergency response. If Nimbi detected signs of physical distress or a potentially dangerous situation, it could share location information, contact emergency services, or alert trusted contacts. The device could also integrate with smart-home systems and daily planning tools to support grocery tracking, routine management, and household safety.

While much of Nimbi's functionality is speculative, the project imagines augmentation as something more ambient and relational than traditional assistive technology. Instead of waiting to be activated, Nimbi would remain constantly present and adapt to each user's unique rhythm, vulnerabilities, and routines; which also means that the design would be customizable to each user, as well.

Process & Critical Making

Nimbi began as a personal question: what would it look like if support systems could physically remain near the people who need them most? No one likes tracking their routines, symptoms, or ailments in an app or journal. Is there a way we could make that easier?

Early versions of the design resembled a realistic fuzzy moth, shiny beetle, or feathery hummingbird. While visually interesting, the design felt uncanny and overly biological. Through iteration, the form shifted toward a softer robotic aesthetic with rounded features and leaf-inspired wings. The redesign made the device feel calmer, more approachable, and more emotionally supportive... something huggable, and less intense.

The prototype process also revealed practical design tensions. A battery-dependent device designed for users with memory impairments risked becoming unreliable simply because it might be forgotten and left uncharged. Integrating solar charging into the wings addressed both usability and sustainability concerns while reinforcing the project's nature-inspired aesthetic, and it would encourage its user to go outside more often than normal.

At the same time, the project raised ethical questions that remain unsolved. The AI systems necessary to make Nimbi adaptive and predictive rely on infrastructures that are environmentally harmful to maintain. Although the device itself incorporates solar-powered technology, the energy consumption associated with contemporary AI processing and data centers would likely outweigh those sustainability efforts. Nimbi therefore exists within an unresolved area of tension: a technology designed to support vulnerable users may also participate in larger systems of environmental harm.

Critical making became central to the project not because the prototype solved every problem, but because the design process exposed the tensions embedded within accessibility technologies themselves.

Accessibility, Usability, & User Experience

Accessibility is not an added feature within Nimbi; it is the foundation of the design itself.

The system prioritizes low-friction interaction through:

- Customizable reminders
- Passive charging
- Adaptable notification systems
- Consistent physical presence
- Emotionally approachable aesthetics

These decisions were made to reduce cognitive load and support users whose conditions already require constant self-monitoring.

At the same time, Nimbi introduces significant usability and ethical tensions. Functionally, it is a surveillance system presented in a companion form. It continuously monitors location, environment, and biometric data unless disabled by the user.

Because of this, trust has become a central user experience issue. Not all users experience surveillance equally, and systems that monitor bodies and behavior can create discomfort, vulnerability, or fear depending on the user's social position and lived experience.

While the current concept includes opt-out controls, future iterations would prioritize more user-controlled permissions, selective monitoring systems, and localized data processing to reduce dependence on continuous cloud-based tracking.

Nimbi therefore exists not only as an accessibility prototype, but also as a reflection on the tradeoffs that often accompany assistive technologies.

Why It Matters

Nimbi explores augmentation not as enhancement for productivity or performance, but as support for vulnerability, embodiment, and everyday survival.

Donna Haraway's concept of the cyborg — a hybrid of organism and machine that disrupts clear boundaries between human and technology — provides an important framework for understanding the project. Nimbi is not simply a tool stored in a pocket or activated when needed; instead, it would follow the user, observe them, learn their habits, and anticipate the users' routines and reminders. Over time, users would likely begin to reorganize their routines, memory practices, and feelings of safety around its presence; and could potentially grow over-reliant on the system.

This raises difficult but necessary questions:

- What happens when cognitive support becomes externalized?

- Can reliance on technological prompts reshape human memory systems?
- Who controls the data generated through accessible technologies?
- When does that assistance become surveillance?

Nimbi does not attempt to offer simple answers to these questions. Instead, the project argues that accessibility technologies are never truly neutral in their design. Every design decision reflects assumptions about care, safety, dependence, privacy, and whose bodies are prioritized within technological systems.

However, the goal of Nimbi isn't perfection, but instead asks that we imagine a more compassionate conversation about what accessible futures could look like when technology is designed around human limitations rather than against it.

Interaction Scenario

At 2:14 PM, Nimbi detects a sudden drop in blood pressure while its user waits alone at a bus stop. The drone lowers its audio output to avoid drawing attention and issues a subtle medication reminder through its user's earbuds. When their vitals continue to decline, Nimbi prepares an emergency contact alert while guiding their user to the nearest seating.

In this moment, augmentation becomes less about efficiency and more about supportive care.

🌱 Intelligent. Compassionate. Sustainable. That's Nimbi. 🌱

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